Design Packet

Benson Lin



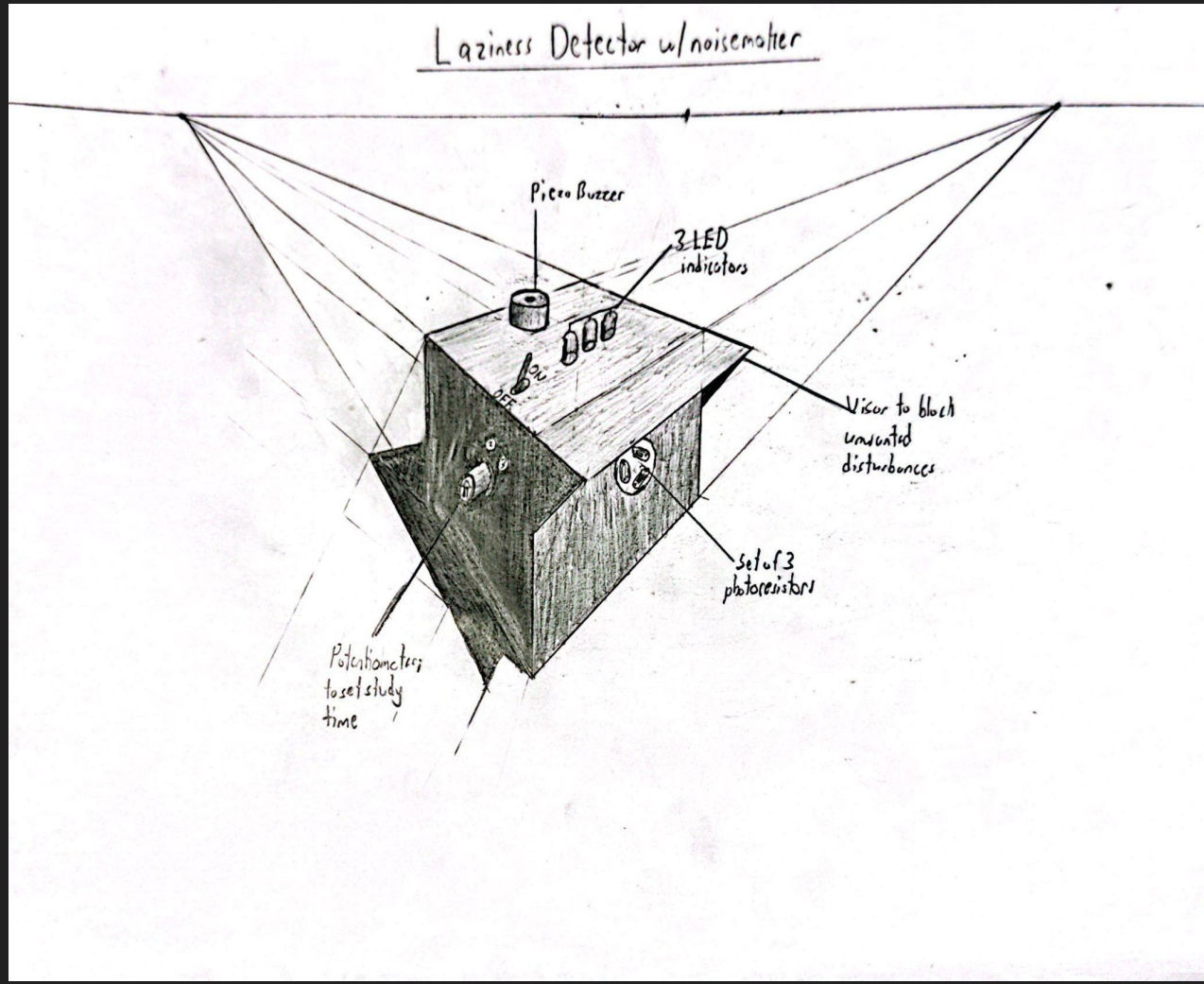

between 3 different study
LEDs by twisting the potentiometer

Summary of Project

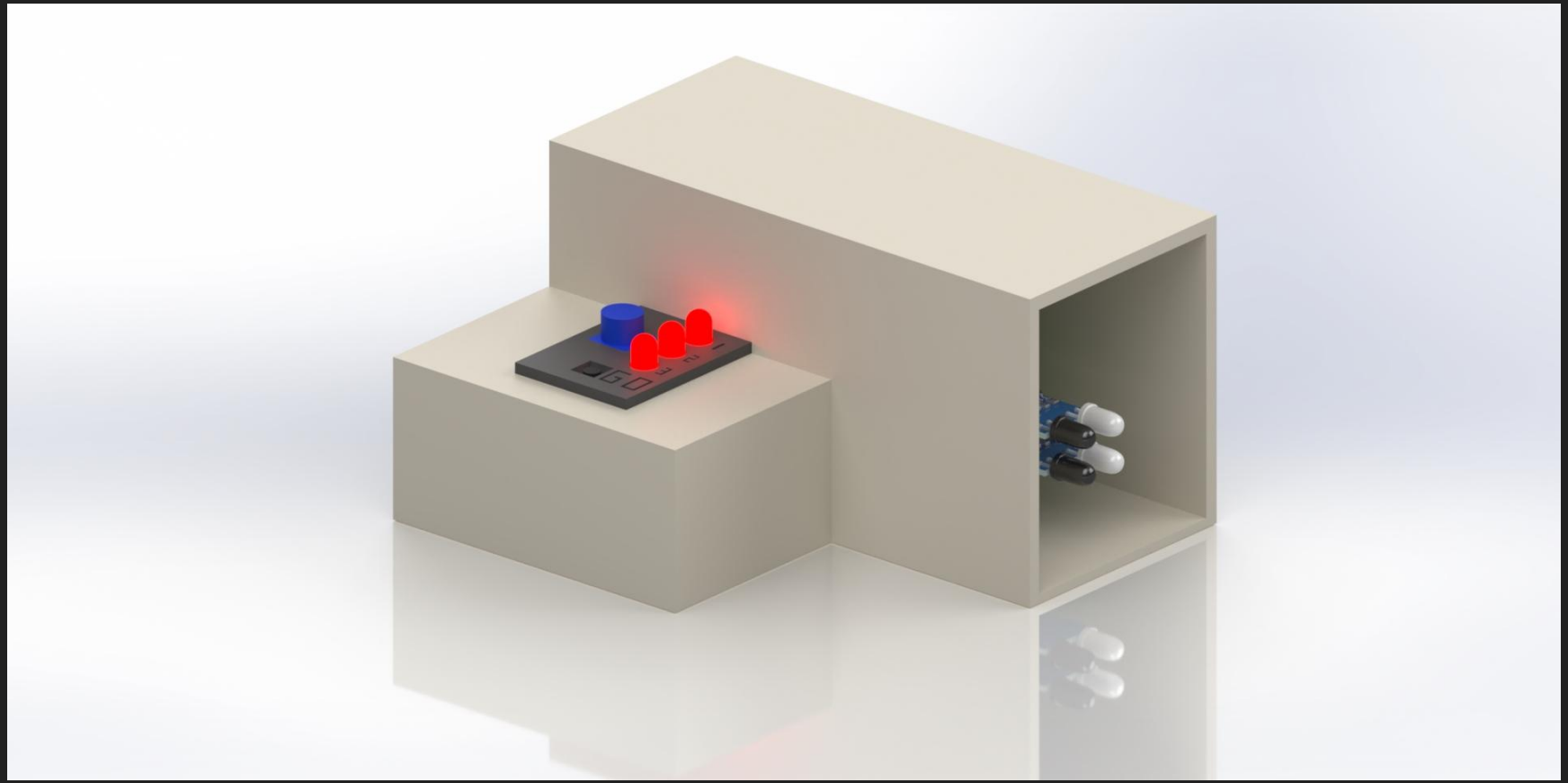
This project was made specifically for people like me who struggle with staying focused while studying, especially before exams. The idea was to create something that forced its user to commit to the task, implementing some form of mild punishment for getting distracted. My device is able to accomplish this task by allowing its user to set a timer for a set amount of study time. Once this timer is started, the user must stay at their workspace until the timer runs out regardless of circumstances. If the user is detected to have left the area, a high frequency noise is emitted until the user is detected to have returned. There were a significant number of issues that arose while designing this device, namely that the original design incorporated photoresistors to detect minute changes in light from the user being near the device. This proved to be difficult, so an infrared distance sensor was implemented instead for a simpler but more effective detection method.

Perspective Sketch

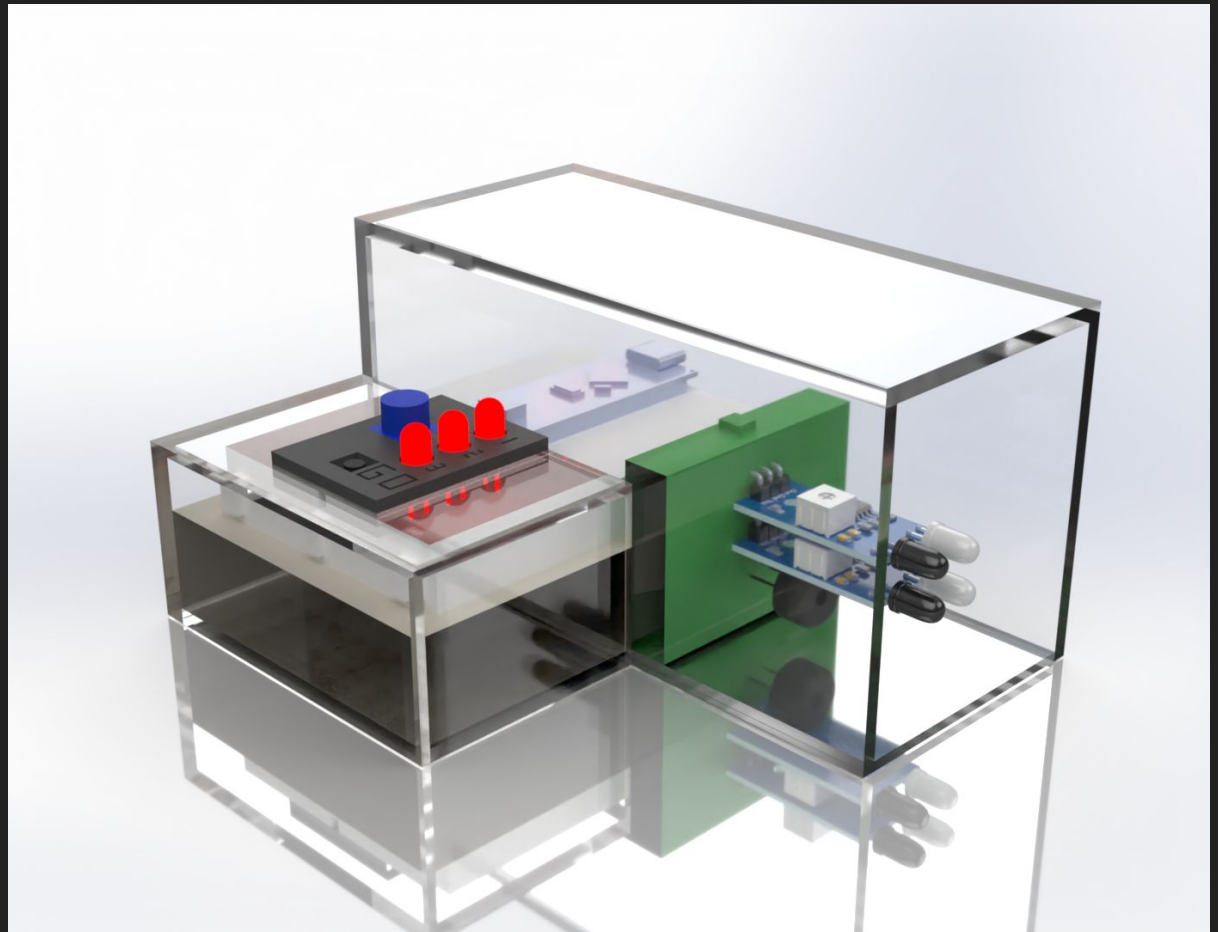
A few things have changed from this original sketch:
-Photoresistors replaced with IR sensors
-Potentiometer relocated to top of device
-Piezo Moved to inside of device
-Cardboard structures changed to better fit the components and for easier fabrication



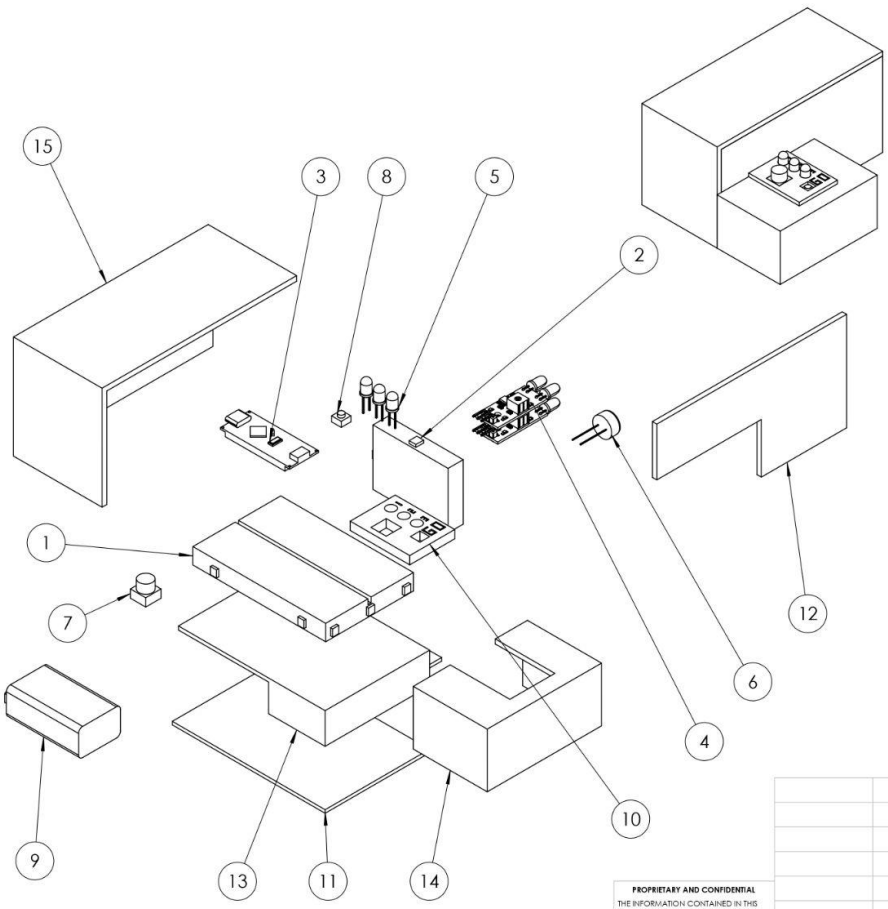
Render of Final Project



Clear Version Render



CAD Drawing Packet



ITEM NO.	DESCRIPTION	QTY.
1	Large Breadboard	1
2	Small Breadboard	1
3	Arduino Nano	1
4	IR sensor	2
5	LED	3
6	Piezo Buzzer	1
7	Potentiometer	1
8	Button	1
9	9V Battery	1
10	3D Printed Part	1
11	Base Plate	1
12	Front Cover	1
13	Cardboard Filling	1
14	Cardboard Shroud 1	1
15	Cardboard Shroud 2	1

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DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
NEXT ASSY | USED ON
APPLICATION | DO NOT SCALE DRAWING

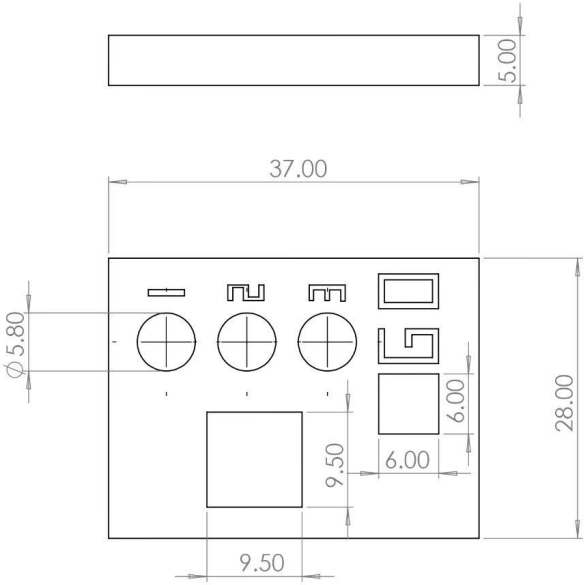
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Q.A.		

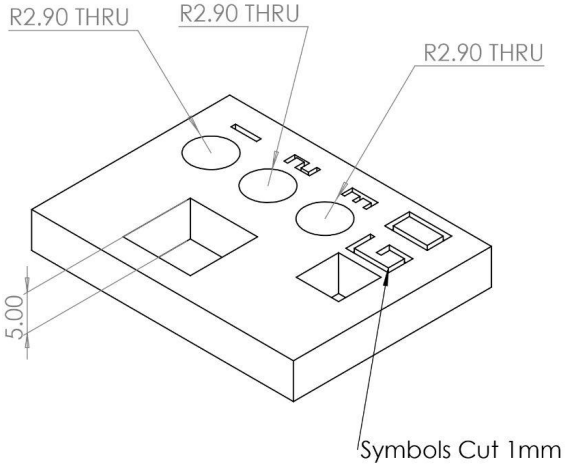
COMMENTS:

TITLE:
Exploded Assembly View

SIZE B | DWG. NO. | REV
SCALE: 1.7 | WEIGHT: | SHEET 1 OF 1

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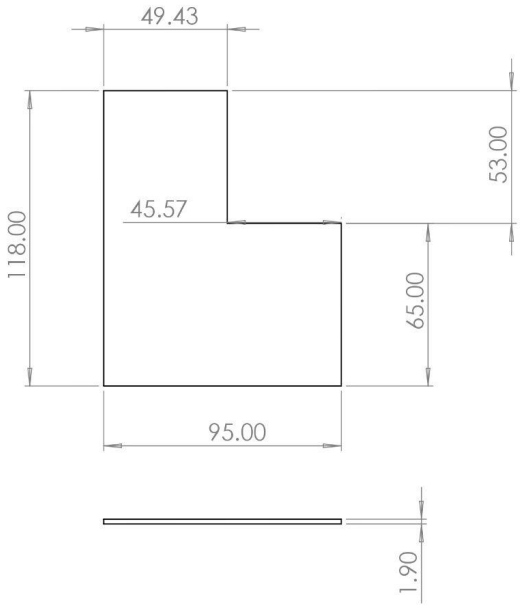
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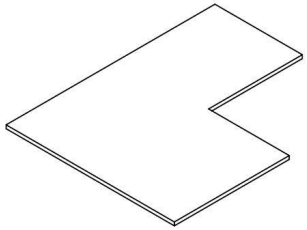
3D Printed Cover

SIZE	DWG. NO.	REV
A		

SCALE: 2:1 | WEIGHT: | SHEET 1 OF 1

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		MATERIAL			
NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

Base Plate

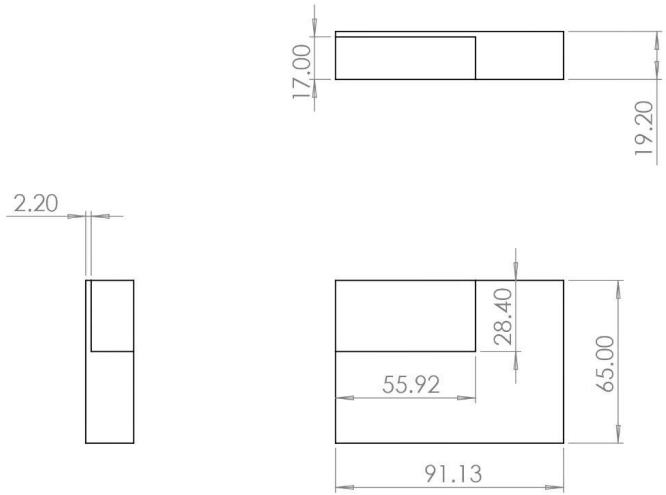
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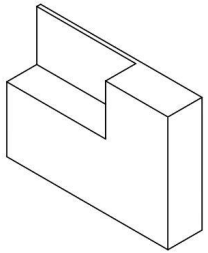
SCALE: 1:2	WEIGHT:	SHEET 1 OF 1

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		MATERIAL	COMMENTS:		
NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

Cardboard Filling

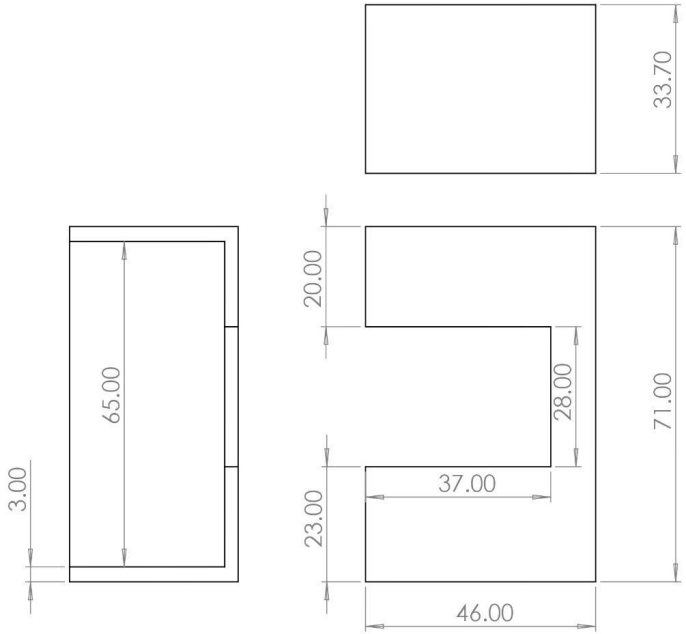
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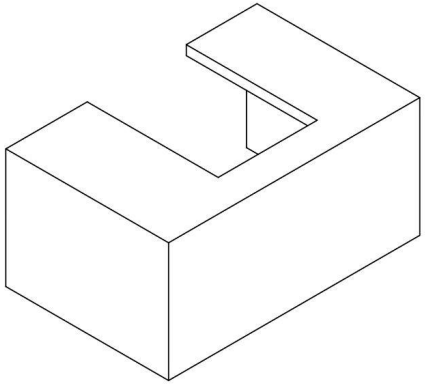
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NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

Cardboard Shroud 1

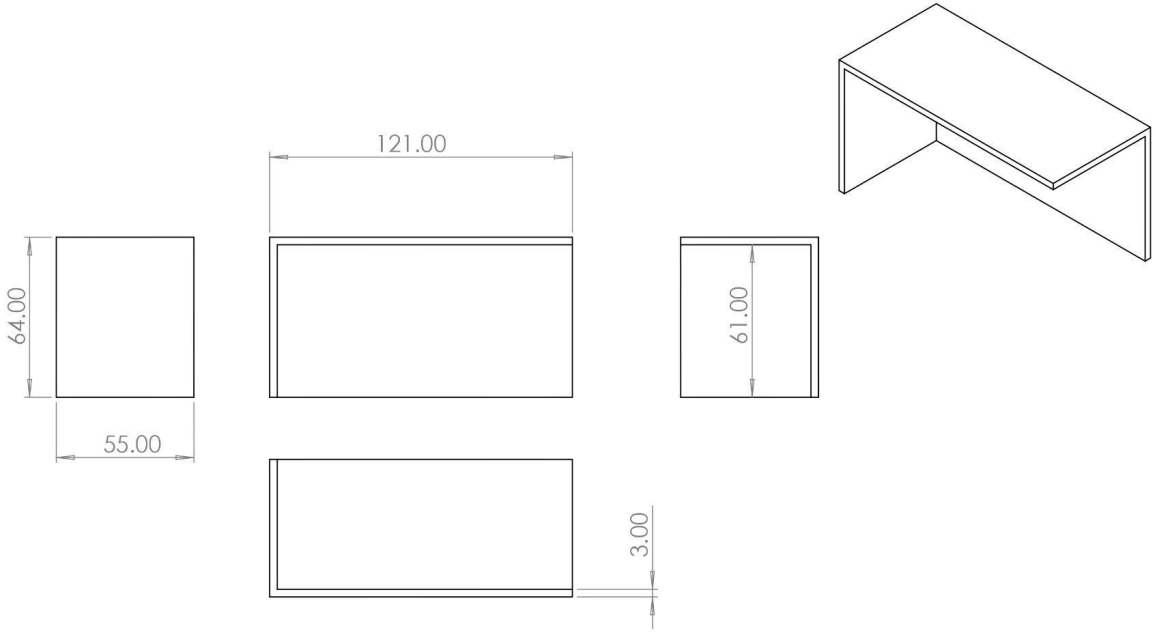
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SCALE: 1:1	WEIGHT:	SHEET 1 OF 1

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APPLICATION		DO NOT SCALE DRAWING			

TITLE:

Cardboard Shroud 2

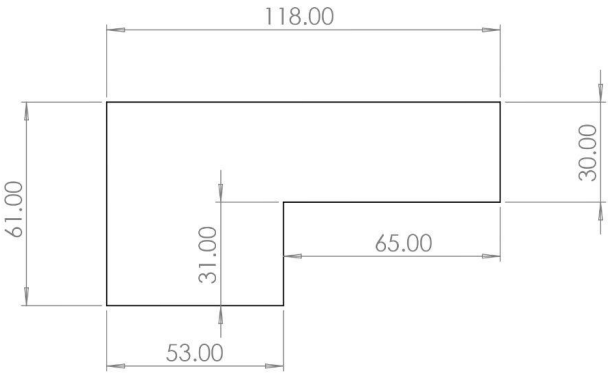
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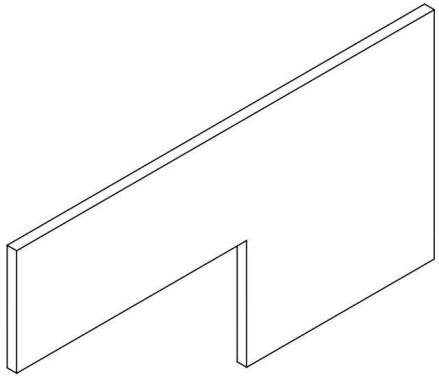
SCALE: 1:2	WEIGHT:	SHEET 1 OF 1

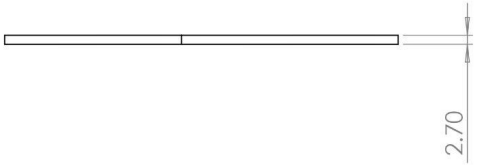
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APPLICATION		DO NOT SCALE DRAWING			

TITLE:

Front Cover

SIZE	DWG. NO.	REV
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SCALE: 1:2	WEIGHT:	SHEET 1 OF 1

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